Fiscal Year:	FY 2013	**Task Last Updated:**	FY 10/21/2012
PI Name:	Shay, Jerry W. Ph.D.		
Project Title:	A Novel Biological Countermeasure and Mitigator of High LET-induced Cancer Progression		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75390-7208	**Congressional District:**	30
Comments:			
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No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	4	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Ugur Eskiocak was a doctoral student who has completed his PhD, Sept 2011. He just left the lab to begin a postdoctoral fellowship in the laboratory of Sean Morrison. Other students are involved in specific aspects of this project (Peter Ly, completing PhD, Nov 2012 and Lu Zhang). Together with Aadil Kaisani (another graduate student) and the recent addition of Mariam El-Ashmawy, we have a solid team of students and postdoctoral fellows (Dr. Sang Kim) assisting in this project. Dr. Roig who became an independent assistant professor investigator moved to Florida and joined a private practice but continues to teach at the University of Miami School of Medicine. Therefore, Dr. Roig is no longer working as a co-investigator on this project.		
COI Name (Institution):			
Grant/Contract No.:	NNX11AC15G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal addresses Radiation Carcinogenesis Cancer Specific Gap 8: Biological Countermeasures (BCM). While progress has been made since NCRP report 132 in 2000, we still do not have the data for assessing the need for BCMs. However, particle emission levels for fatal cancer risks are projected to be violated for a Mars mission under all possible scenarios. With current permissive exposure limits most astronauts would be limited to about 6-8 months on deep space missions. Thus, if a goal of NASA is to have experienced crews fly multiple times, it is prudent to assume that even with new research to modify uncertainty reductions in fatal cancer risk projection models, that the development of BCMs should now be initiated. The work in progress has been to investigate, Bardoxylone methyl (BARD), an oral available non toxic anti-oxidant inflammatory modulator that is currently in Phase IIB clinical trials for patients with renal insufficiency. BARD is a synthetic triterpenoid and there is a significant molecular understanding of its mode of action. We have already completed investigations with low and high LET irradiation in both normal human lung and colonic epithelial cells and have shown that 10-50nM BARD is an effective radiation BCM, reducing double-strand DNA damage and anchorage-independent cancer progression. Based on our results in human epithelial cells, we now propose to demonstrate BARD is effective in mouse models of cancer. We have a commitment from Reata Pharmaceuticals (Irving, TX) to provide the compound required for animal testing. We believe BARD has an excellent chance of being developed into a potent BCM to protect against high LET irradiation. There are data showing that BARD can protect against radiation-induced mucositis in a hamster cheek pouch assay. In addition, BARD protects against radiation-induced intestinal damage in zebrafish development. Lastly, we have evidence that BARD can mitigate gamma-irradiation (using the colony formation assay) if provided within 30 minutes after IR. In summary, the main objective of this V-NSCOR application is to investigate if BARD is a potent BCM and/or mitigator in mouse models that are predisposed to cancer. Specific aim 1: To use a lung cancer susceptible mouse model that develops significantly more invasive lung tumors when high LET is provided in fractionated/protracted doses over 5 to 10 days (e.g 0.2 Gy x 5 days or 0.1 Gy x 10 days 56Fe 1GeV, compared to single acute 1 Gy 56Fe). Our objective is to determine if pretreatment with BARD prior to acute or fractionated IR reduces the number of invasive (e.g. more lethal) cancers. We will examine at least 3 different ions. Specific aim 2: To test BARD as a BCM in mouse models exposed to SPE simulations. As part of the UT Southwestern Lung NSCOR and the Georgetown GI NSCOR, we have lung cancer and colon cancer susceptible mice that have been treated with solar particle simulations. We propose to test if providing BARD to these animals before irradiation reduces the incidence or progression of cancer. Specific aim 3: To test BARD as a radiation mitigator by providing it to mice within 60 minutes after irradiation. We will compare SPE simulations (total body exposure 2 Gy Protons) and at least two other ions (56Fe at 600 meV, LET 175 and 28Si at 300 MeV LET 70). The significance of the proposed work is that we will be able to demonstrate if BARD reduces cancer incidence or progression in two important tissues (lung and colon) that are susceptible to radiation-associated cancer using doses and doses rates recommended by NASA. This research should provide new data to help reduce uncertainties in current risk projection models.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project may lead to the development of a novel radiation countermeasure that will not only be important to protect astronauts from the effects of space radiation, but may also have utility in protecting normal human tissues when radiotherapy is used to treat cancer patients.
	This project has been to investigate bardoxolone methyl (BARD, also called CDDO-Me) and an analogue (CDDO-EA), both oral available non-toxic anti-oxidant inflammatory modulators (AIMs) as potential radiation countermeasures/mitigators. BARD is a synthetic triterpenoid and there is a significant molecular understanding of its mode of action. We have already completed a series of investigations with low-LET and high-LET irradiation in both normal human lung and colonic epithelial cells and have shown that single doses of 10-50nM BARD/CDDO-Me are effective radiation biological countermeasures, reducing double-strand DNA damage and anchorage-independent cancer progression. We also have experimental evidence that BARD can mitigate steps in cancer progression from gamma-irradiation (using colony formation assays) if provided 0.5-12 hours after irradiation via a novel mechanism of action that involve enhanced DNA damage repair. Based on these encouraging results in human epithelial cells, we have initiated experiments to demonstrate if an ethyl amide analogue of BARD (CDDO-EA) is an effective countermeasure against high-LET irradiation in mouse lung and colon cancer susceptible models. We believe CDDO-Me/CDDO-EA has an excellent chance of being developed into a potent biological countermeasure to protect astronauts against high-LET irradiation and possibly to serve as a mitigator after exposure to radiation. All criteria for approval by the FDA for BARD as a biological countermeasure can be met based on current data except for protection from high-LET irradiation in an animal model. In summary, the main objective of this project is to investigate if CDDO-EA is a potent biological countermeasure and/or mitigator of SPE simulations and NASA priority HZE ions in mouse models that are predisposed to solid cancers. Since BARD is an anti-inflammatory modulator of oxidative damage, we believe BARD may also protect against HZE induced short-term and long-term neurodegenerative cognitive deficits and perhaps cardiovascular injury. We have already initiated research at the NSRL that support approaches to moving BARD and CDDO-EA forward as lead space radiation biological countermeasure candidates. Our hypothesis is that BARD/CDDO-EA may lower the HZE carcinogenic risks. Our proposed research is hypothesis based, focuses on solid cancer risks, and is mechanistically driven. Specific Aim 1: To use the LA-1 lung cancer susceptible mouse model that develops significantly more invasive lung tumors when high-LET is provided in fractionated/protracted doses. Our objective is to determine if pretreatment with non toxic doses of CDDO-EA prior to acute or fractionated HZE irradiation reduces the number of invasive (e.g. more lethal) lung cancers. We will compare this to other priority HZE ions using fluences and energies that are representative of GCR including 28Si and 16O. We will also test a mouse model of colon cancer (CPC;APC). Specific aim 2: To test CDDO-EA as a biological countermeasure in mouse models exposed to SPE simulations. Specific aim 3: To test CDDO-EA as a radiation mitigator by providing it to mice within 30-60 minutes after irradiation (via IP and orally administered gavage). We will compare SPE simulations (total body exposure 2 Gy protons) and two

Task Progress:	other ions (56Fe at 600 MeV, LET 175 at dose ranges between 0.1 Gy and 0.5 Gy and 28Si at 300 MeV LET 70 at doses between 0.2 and 0.8 Gy). We will test single acute versus fractionated doses of 56Fe and 28Si. Summary of Progress: 1. SPE (a solar particle event simulation) irradiation decreases the lifespan of LA1 K-ras lung cancer susceptible mice and results in an increase incidence of carcinoma. This is similar to what is observed in animals irradiated with fractionated Fe- 2. An increase in expression of certain inflammatory factors is observed in mice exposed to SPE 70 days post-irradiation well before the development of more lethal invasive cancer 3. LA1 K-ras mice on CDDO-EA diet during fractionated Fe- irradiation exhibit a decreased incidence in invasive carcinoma in comparison to mice on control diet 4. LA1 K-ras mice fed CDDO-EA without irradiation show reduced tumor progression 5. CPC;APC colon cancer susceptible mice exposed to a SPE simulation show a 2-fold increase incidence of polyps (benign tumors) and a 3-fold increase incidence of invasive carcinomas 6. Providing CDDO-EA in lab chow 2-3 days prior to SPE simulation results in a slight decrease in the incidence of polyp formation but a dramatic (almost 3-fold) decrease in the more lethal invasive carcinomas 7. CDDO-EA provided prior to Fe or SPE irradiation results in overall survival increases over control animals in both the LA-1 lung cancer mouse model and the CPC;APC colon cancer mouse model 8. Providing CDDO-EA to wild type mice results in increased phosphor Nrf-2 in the GI tract within 24 hours peaking at 48 hours and also reduces basal oxidative stress in the GI tract 9. Providing CDDO-EA in lab chow three days prior to a lethal doses (7.5 to 10 Gy) of ionizing radiation resulted in 50% survival of wild type mice 10. Providing CDDO-EA in lab chow to young LA-1 lung cancer susceptible mice for 100 days greatly reduces the size and progression of tumors.
Bibliography Type:	Description: (Last Updated: 11/21/2025)
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